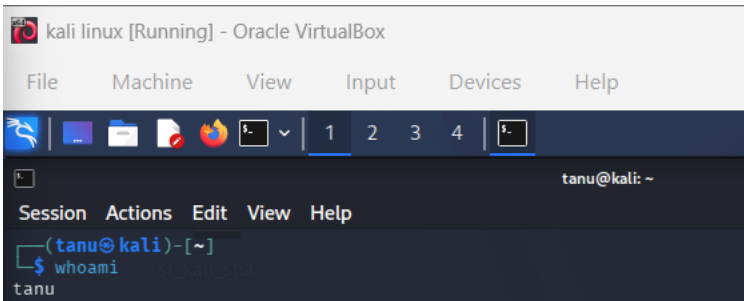


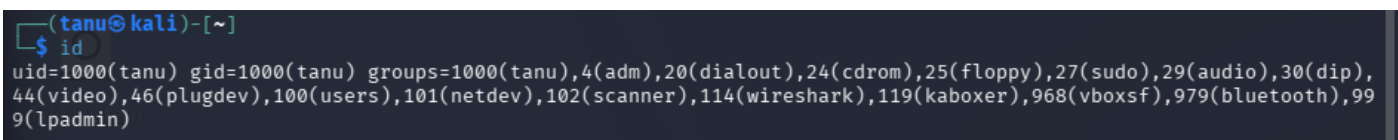
Part A — User and System Investigation

1. Determine the username of the account currently being used



```
kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
tanu@kali: ~
Session Actions Edit View Help
(tanu@kali)-[~]
$ whoami
tanu
```

2. Display UID, Primary GID, and Supplementary Groups



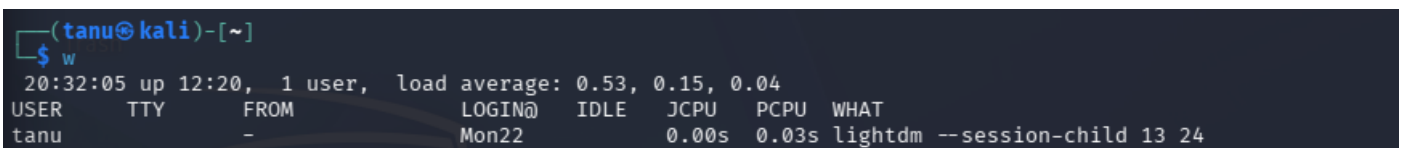
```
(tanu@kali)-[~]
$ id
uid=1000(tanu) gid=1000(tanu) groups=1000(tanu),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),
44(video),46(plugdev),100(users),101(netdev),102(scanner),114(wireshark),119(kaboxer),968(vboxsf),979(bluetooth),99
9(lpadmin)
```

3. Display all users currently logged in to the system



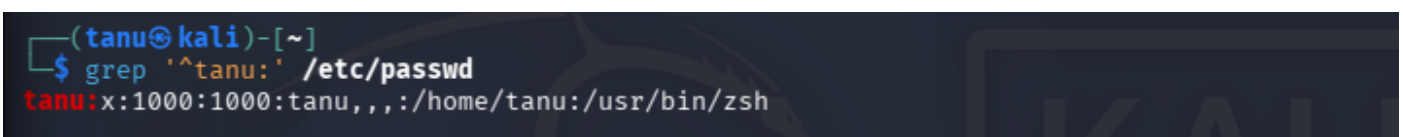
```
(tanu@kali)-[~]
$ who
tanu      seat0      2026-08-31 22:26 (:0)
```

4. Determine what activities the currently logged-in users are performing



```
(tanu@kali)-[~]
$ w
20:32:05 up 12:20, 1 user, load average: 0.53, 0.15, 0.04
USER      TTY      FROM          LOGIN@      IDLE        JCPU      PCPU      WHAT
tanu      -            Mon22                0.00s      0.03s      lightdm --session-child 13 24
```

5. Examine /etc/passwd and identify information for the current account



```
(tanu@kali)-[~]
$ grep '^tanu:' /etc/passwd
tanu:x:1000:1000:tanu,,,:/home/tanu:/usr/bin/zsh
```

6. Determine how many user-account entries are present in /etc/passwd



```
(tanu@kali)-[~]
$ wc -l /etc/passwd
57 /etc/passwd
```

7. Display only the username and UID fields from /etc/passwd

```
(tanu@kali)-[~]
$ awk -F: '{print $1, $3}' /etc/passwd
root 0
daemon 1
bin 2
sys 3
sync 4
games 5
man 6
lp 7
mail 8
news 9
uucp 10
proxy 13
www-data 33
backup 34
list 38
irc 39
_apt 42
nobody 65534
systemd-network 997
_dhcpd 995
tss 989
strongswan 100
systemd-timesync 988
_gophish 101
iodine 102
messagebus 987
tcpdump 986
```

8. Find all accounts whose UID is 0

```
(tanu@kali)-[~]
$ awk -F: '$3 == 0 {print $1, $3}' /etc/passwd
root 0
```

9. Compare /etc/passwd and /etc/shadow

```
(tanu@kali)-[~]
$ ls -l /etc/passwd /etc/shadow
-rw-r--r-- 1 root root 3263 Jul 31 18:50 /etc/passwd
-rw-r----- 1 root shadow 1464 Jul 31 18:48 /etc/shadow

(tanu@kali)-[~]
$ sudo head -5 /etc/passwd
[sudo] password for tanu:
root:x:0:0:root:/root:/usr/bin/zsh
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
```

Part B — Creating and Verifying Users

1. Create the users alice, bob, and charlie

```
(tanu@kali)-[~]
$ sudo useradd -m alice

(tanu@kali)-[~]
$ sudo useradd -m bob

(tanu@kali)-[~]
$ sudo useradd -m charlie
```

Part B — Creating and Verifying Users

2. Verify that all three accounts have been added to /etc/passwd

```
(tanu@kali)-[~]
$ grep -E '^(alice|bob|charlie):' /etc/passwd
alice:x:1001:1001::/home/alice:/bin/sh
bob:x:1002:1002::/home/bob:/bin/sh
charlie:x:1003:1003::/home/charlie:/bin/sh
```

3. Determine the UID assigned to each newly created user

```
(tanu@kali)-[~]
$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice)

(tanu@kali)-[~]
$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob)

(tanu@kali)-[~]
$ id charlie
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie)
```

4. Determine the home directory and default shell assigned to alice

```
(tanu@kali)-[~]
$ grep '^alice:' /etc/passwd
alice:x:1001:1001::/home/alice:/bin/sh
```

5. Inspect /home and verify that the required home directories were created

```
(tanu@kali)-[~]
$ ls -l /home
total 16
drwx----- 5 alice  alice  4096 Sep  1 20:59 alice
drwx----- 5 bob    bob    4096 Sep  1 20:59 bob
drwx----- 5 charlie charlie 4096 Sep  1 20:59 charlie
drwx----- 20 tanu   tanu   4096 Sep  1 20:26 tanu
```

6. Set passwords for all three users

We need to set a password for each account.

```
(tanu@kali)-[~]
$ sudo passwd alice
[sudo] password for tanu:
New password:
Retype new password:
passwd: password updated successfully

(tanu@kali)-[~]
$ sudo passwd bob
New password:
Retype new password:
passwd: password updated successfully

(tanu@kali)-[~]
$ sudo passwd charlie
New password:
Retype new password:
passwd: password updated successfully

(tanu@kali)-[~]
$
```

7. Switch from the current account to alice using a login shell

```
(tanu@kali)-[~]  
$ su - alice  
Password:  
$  
$ whoami  
alice  
$ exit
```

Part C — Group Management and Access Control

```
(tanu@kali)-[~]  
$ sudo groupadd security  
sudo groupadd developers  
[sudo] password for tanu:
```

Part C — Group Management and Access Control

2. Add alice and charlie to the security group

```
(tanu@kali)-[~]  
$ sudo usermod -aG security alice  
sudo usermod -aG security charlie
```

3. Add bob to the developers group

```
(tanu@kali)-[~]  
$ sudo usermod -aG developers bob
```

4. Verify the group membership of all three users

```
(tanu@kali)-[~]  
$ id alice  
id bob  
id charlie  
uid=1001(alice) gid=1001(alice) groups=1001(alice),1004(security)  
uid=1002(bob) gid=1002(bob) groups=1002(bob),1005(developers)  
uid=1003(charlie) gid=1003(charlie) groups=1003(charlie),1004(security)
```

5. Display the entries for security and developers from /etc/group

```
(tanu@kali)-[~]  
$ grep -E '^((security|developers):)' /etc/group  
security:x:1004:alice,charlie  
developers:x:1005:bob
```

6. Add bob to the security group without removing him from developers

```
(tanu@kali)-[~]  
$ sudo usermod -aG security bob
```

7. Verify that bob now belongs to both groups

```
(tanu@kali)-[~]  
$ id bob  
uid=1002(bob) gid=1002(bob) groups=1002(bob),1004(security),1005(developers)
```